

Situated Visualization of IIoT Data on the Hololens 2

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ABSTRACT

Augmented Reality for the Industrial Internet of Things (AIIT) is a use case in the larger research project *Immersive Media Lab* [2], which is currently being conducted at the University of Applied Sciences St.Pölten. This use case deals with the challenges of AR data visualization on AR HMDs in smart factories using the example of an industrial furnace. The application is intended to support people in two ways: On the one hand, new personnel should be able to quickly and easily understand the processes happening in the plant, and on the other hand, conditions for maintenance work that are invisible from the outside should be displayed in an understandable and easily consumable way directly situated at the system using AR. In addition to conventional displays (dashboards) for the representation of abstract sensor values, we use the concept of *Indexical Visualization*, which can reduce the cognitive load for the viewer. The system itself is expanded in AR in such a way that the information to be displayed becomes a natural part of the scene.

In this paper, we describe the steps necessary to create a mixed reality industrial application for the Microsoft Hololens 2 in terms of: optimizing a digital 3D model of the facility, establishing the data interface to the plant using an OPC-UA to MQTT bridge, correctly aligning the large 3D object in the real environment, and visual metaphors for making the machine data understandable for users.

Index Terms: Human-centered computing—Visualization—Visualization application domains—Information visualization; Human-centered computing—Human-computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality

1 INTRODUCTION

The benefits of using Augmented Reality (AR) in industrial applications to support workers in tasks like training [9, 32], maintenance-assembly-repair [11, 12], product inspection [17], facility monitoring [22, 23], human-robot-collaboration [16] and remote assistance [6, 21, 31] have been out pointed many times [5, 7, 26]. Smartphones are currently the prevalent AR platform (video-see-through, handheld AR). AR with transparent screens (optical-see-through, glass-based AR) using head-mounted displays (HMDs) has significantly higher requirements for the necessary hardware and ergonomic aspects and is therefore currently still a niche application. Prototypes for such devices have been around since as early as 1968 [27], but it is only now that technological capabilities have caught up with the requirements to use these devices in not only experimental settings, but also in professional environments (e.g. in aerospace manufacturing [1]). Also, the current expenditures of the major IT players in this area for the further development of this platform are considerable, so that there is a realistic chance that glass-based AR devices will become as widespread as smartphones are today [3].

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Figure 1: An industrial aluminum melting furnace was chosen to demonstrate the situated visualizations of IIoT data using AR. As almost all of the states of the control system are invisible from the outside, AR is well suited for this task in supporting workers for learning about and monitoring the system.

In manufacturing, great efforts are currently being made to transform classic production plants into *cyber-physical systems* [28, 33]. The concept of *cyber-physical equivalence* assumes that real plants have a digital twin that is updated with data from reality using *Industrial Internet-of-Things (IIoT)* protocols [10]. Smart factories are networked and data-driven to the highest degree, with humans increasingly performing a control function rather than a steering role as a result of advancing automation. It is therefore even more important to be able to transmit the virtually available data in real time back to the human operator. AR HMDs are particularly well suited for this purpose, as virtual readouts from data points can be anchored directly where they are needed in real space. In addition to the direct positioning of data displays as dashboards, AR can also be used to represent more complex situations that are based on multiple data points, directly as natural, virtual extensions of the real object structure. The goal is a reduction of the interpretation effort for humans about the overall state of a system, which results from many individual data points and can thus also be conveyed passively and continuously.

The use case presented here describes an application scenario of AR HMDs in an industrial context to show possibilities how this technology can be used to transmit and visualize machine data for employees in a meaningful way and how challenges regarding application and interface development can be solved.

2 AR AS A BRIDGE BETWEEN THE HUMAN AND IIoT

A goal of the use case was to work with an industrial facility to get real data as well as input and feedback from the personnel. We partnered with an aluminum production plant and performed interviews to identify possible applications where AR technology could be supportive. Two scenarios were identified through discussions with

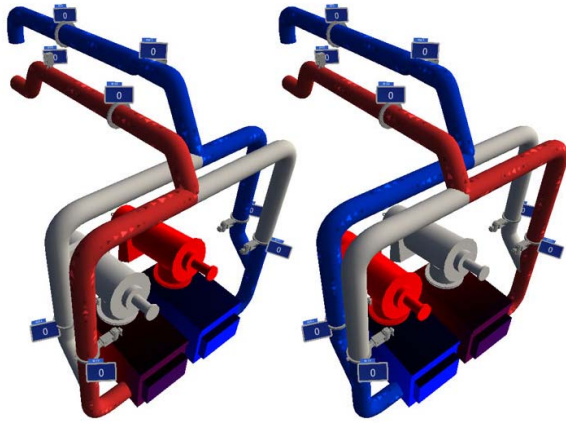


Figure 2: The combustion system has two gas burners which alternate in operation. A common system of air pipes is configured differently regarding in- and outlet air, depending on the burner active. However, for production workers this state is observable only through isolated sensor values.

the responsible technicians in which the properties of AR HMDs can be advantageous to the operating personnel:

- In learning about the operation of a plant that is unfamiliar to the person: It is often the case that plant technicians with specialized knowledge are called to an industrial plant and are expected to get to know it as quickly as possible. Any assistance that can speed up this process and provide educational support is deemed positive in this regard.
- When inspecting and maintaining the plant directly on site: The additional virtual information layer facilitates the inspection of sensor values, which are otherwise usually communicated via speech connection to an employee in the control room. Glass-based AR promises the great advantage of being able to present the relevant information in a situated manner without blocking the hands of their user.

2.1 Use Case: Industrial furnace

Based on these tasks, an industrial furnace (see Fig. 1) for aluminum melting was selected, on which the challenges of using AR in an industrial context can be demonstrated. The furnace's combustion system consists of two alternately active gas burners, whereby a configurable network of pipes for supply and exhaust air must be controlled depending on the burners activities. The pipes transport either supply air or exhaust air, depending on which burner is active (c.f. Fig. 2). This concept is easy to represent visually, but is currently presented to technicians exclusively via a large number of isolated sensor values in graphs. This abstraction requires a high level of human interpretation in order to understand the current state of the machine. Furthermore, the displayed values are not visibly located at the plant, so that they have to be read from abstract technical drawings when getting to know the system. Therefore, this learning and interpretation process to create understanding is time-consuming and error-prone.

2.2 Application design

To support the staff using AR, two different scenarios were implemented in our application: The display of a miniature model and direct visualizations on the furnace. In both scenarios, raw machine data sourced from IIoT protocols can be displayed directly as abstract values with graphs and widgets. Furthermore, ways to provide

an easier-to-understand overview of the overall state of the plant using the concept of *Indexical Visualization* were explored and will be elaborated in Sect. 6.

- **Miniature Model:** The primary goal of this mode is to help people become familiar with the overall system. In this mode, the furnace is visualized three-dimensionally as a miniature model. Essential components are provided with verbal annotations and animations of the air flows support the understanding of the workings of the whole system. It is possible to display a simulation as well as actual live machine data.
- **Visualizations directly at the furnace:** The visualization of machine data directly at the furnace should support technicians especially during inspections and maintenance.

As a platform for the implementation, we decided to use the Microsoft HoloLens 2 as a glasses-based AR HMD, since at the time of project implementation it offered the most mature technology in terms of spatial tracking and visualization on see-through displays, as well as a well-developed SDK with the MRTK2. Furthermore, the HoloLens 2 is certified with ANSI Z87.1, CSA Z94.3 and EN 166 for impact protection and hence suitable for industrial applications [20]. A particular challenge for this platform is the mobile computing hardware used by the device, which is limited compared to conventional PCs and has therefore tighter requirements regarding real-time visualizations. Unity¹ in version 2019.4 with the MRTK version 2.5² were used for application development.

For both use cases, a number of challenges need to be solved in terms of asset optimization, spatial location of larger-than-life objects, interface to machine data, and data visualization, which are described below.

3 AUTOMATIZED VS. MANUAL 3D MODEL OPTIMIZATION

To meet the set goals (miniature model and mixed reality overlay), it is necessary to create the machine as an optimized 3D model for real-time display on the AR glasses. In our case, we had a precise CAD (computer-aided design) model of the furnace available, but due to its size and complexity, it had to be optimized for use in the AR application. There are numerous approaches for this, with the reduction of vertices and omission of unnecessary details playing a major role. A manual redesign under these aspects is rarely an option in many cases due to the large amount of work involved with such assets, which is why specialized software (e.g. Okino Nugra³) exists for this task. However, the use of such kind of software did not prove to be effective in our case, as experiments with this kind of software were not satisfactory with respect to the data reduction heuristic it applied. We therefore opted for a manual reconstruction, which in our case was completed by a capable 3D modeler within a few days and reduced the original 6 million to around 80 thousand polygons.

Regarding the optimization of the 3D model, the same methods can be applied as used in conventional real-time 3D applications: Reduction of vertices, application of simple shaders for surface shaping and intensive use of geometry instancing [29]. The latter is a method that is beneficial for many industrial facilities, where many copies of identical parts exist at different locations and therefore promise large performance increases (it also brought significant improvements for the rendering time in our case).

¹<https://unity.com/>

²<https://microsoft.github.io/MixedRealityToolkit-Unity/>

³<https://www.okino.com/nrs/nrs.htm>

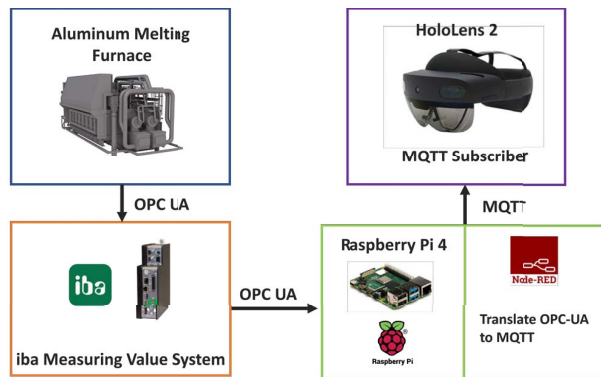


Figure 3: OPC-UA data is translated to MQTT topics using Node-RED, as MQTT is lightweight and can easily be consumed on the Hololens 2.

4 ALIGNING LARGE MODELS WITH ITS REAL COUNTERPART

A fundamental challenge is the spatially correct representation of virtual objects over a real facility. Quick and correct alignment is a prerequisite for any further visualization steps that follow. The Hololens 2 automatically sets its own global coordinate system when the application is launched, so other virtual objects must first be placed with respect to this reference.

Partially automated methods for pose finding of virtual objects in reality exist, which are capable of creating a pose reference using 2D fiducials (e.g. ArUco markers [8]) or using 3D object tracking [30]. However, these methods are usually applied to objects that are in general much smaller than an industrial plant. In our use case, however, the real object and its virtual twin are significantly larger than that, so that minimal angular deviations in the registration of a marker at the distance of 10 meters already leads to significant deviations. We therefore decided to use a manual method that is similar to the process described by [1]. It is quickly applicable for the correct local registration of the 3D model, by simply defining 3 points in space. From these, two vectors and furthermore a coordinate frame (object pose) can be derived using an orthonormalization procedure. The same 3 points are first defined in the virtual model and then tagged by the user in the application at the corresponding positions in real space using a hand gesture. Now the pose of the real object can be calculated using the same method and then the orientation of the virtual model can be adjusted accordingly to the real positions. This localization is done once at the beginning of the application. In informal tests, the local tracking of the Hololens 2 proved to be sufficiently stable: Slight drift in the centimeter range was observed, but was not perceived as disturbing.

5 BRINGING IIOT DATA TO THE HOLOLENS 2

In order to be able to consume the machine data on the Hololens, an interface to the industrial network had to be established. The control and sensor technology of the plant communicates with the OPC-UA (Open Platform Communications - Unified Architecture) standard [15] and is aggregated with a process data management system⁴. Theoretically, this data could also be consumed directly on the Hololens 2 via this interface, but at the time of project implementation, no executable and in our context feasible library for processing OPC-UA data on the Hololens 2 with Unity3D was available. A proxy application was therefore developed based on our findings in [13], which automatically translates OPC-UA data points into MQTT (Message Queuing Telemetry Transport) topics and can thus

⁴IBA-PDA V6. <https://www.iba-ag.com/de/ibapda>

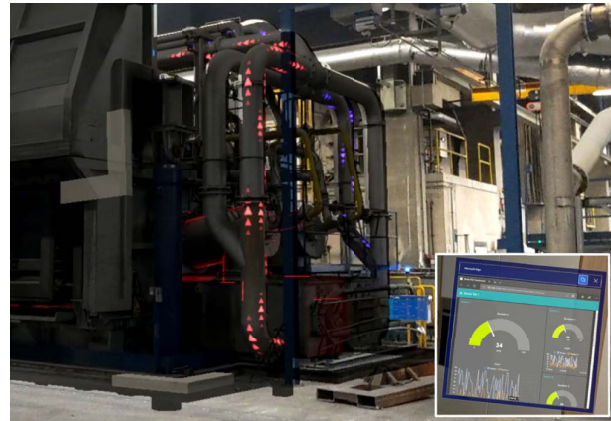


Figure 4: Indexical Visualization of the air flow and active gas burners as naturally behaving overlays on the real machine. Furthermore, classic symbolical displays like gauges and graphs can be freely placed.

be easily consumed by arbitrary subscribers. The MQTT protocol is very resource-efficient and therefore can be processed well on devices with limited computing power such as the Hololens 2. Also it has been proven to work well in industrial environments [14]. The proxy application was implemented using Node-RED [25], using an OPC-UA module extension [4] and runs on a RaspberryPi 4 (c.f. Fig. 3). Although this system is inexpensive, the flexibility and fast development cycles associated with implementing a separate proxy this way proved to be beneficial in the development process.

6 DATA VISUALIZATION

Our application has two modes of operation (miniature model and visualization directly on the furnace), which aim at different purposes (learning and situated visualization) but have a lot in common regarding the representation of IIoT data. For both, the raw display of sensor values as gauges at the point where they are measured is possible. The added value of these displays comes from the fact that they can be freely placed and, if required, clearly summarized as dashboards. The conventional visualization of data as symbolic or iconic displays such as gauges, digits, graphs, etc. can be realized quickly and the corresponding software modules can be easily reused for a wide variety of measured variables. However, the interpretive performance of the data itself is left to humans.

6.1 Indexical Visualization using AR

Inspired by the concept of *Indexical Visualization*, which was formalized by Offenhuber and Telhan [24], we therefore additionally developed an alternative visual representation. Central points in this concept are the *framing of the underlying object in such a way that it becomes legible*, as well as that the *indexical display remains linked with the underlying phenomenon through its visual appearance*. The visual extension of an industrial plant by indexical properties instead of symbolic or iconic displays (like gauges) aims at a modification of the object's appearance in a way that the whole machine is turned into a display of the inherent processes. While Offenhuber and Telhan's analysis focuses primarily on physical displays, AR appears to be a natural platform for modifying and extending real objects based on existing virtual data.

A control system such as the gas burner air circuit represents an overall state, but it is represented by many individual sensors and actuators. Their isolated consideration requires knowledge of the system as well as experience regarding their dependencies in order to be able to interpret the overall state. By extending the system

with an indexical display, the personnel can be supported in this interpretation process that the cognitive effort is minimized and passive monitoring without value interpretation is made possible. In our case, we decided to use a situated overlay of the pipes and combustion chambers, in which the air flow and burner activity is directly superimposed on the overall system of the plant by means of coloring of components and animated arrows (c.f. Fig. 4). Due to the currently present COVID pandemic and restrictions with access to the facilities we were not yet able to perform thorough user tests about the validity of our assumptions regarding the reduction of cognitive efforts. However, informal feedback we got from the factory technicians support our hypothesis of the usefulness of these kinds of visualizations.

7 CONCLUSION AND OUTLOOK

In this paper we presented a use-case for the application of AR HMDs for the situated visualization of IIoT data of an industrial furnace. It was shown which preconditions have to be established in order to precisely spatially anchor and display a virtual 3D model of the plant over the real object using the Hololens 2. A major challenge here is the optimization of existing CAD models, which are not directly suitable for visualization on mobile AR HMDs due to the level of detail. In our case, manual remodeling proved to be more expedient, although this is probably not practical for larger plants. In the future, however, this problem could be simplified using remote rendering [19], where the computation of the visualizations is done on powerful external computers and the matching image is transmitted as a video stream to the end device. Also with regard to spatial alignment, technologies such as cloud or spatial anchors [18] based on visual features of the environment promise that this manual step will no longer be necessary in the future. For the transfer of data available in the OPC-UA protocol to the Hololens 2, a proxy for translation to the MQTT protocol using Node-RED proved useful and quickly adaptable. Since a variety of other protocols are supported in the Node-RED ecosystem, this architecture could also be adapted to other IIoT protocols (e.g. CoAP, AMQP, DDS). Regarding the visualization of the machine data, we distinguish between direct symbolic or iconic representation of the data (as classical value displays) and indexical representation as a virtual extension of the real object. The hypothesis that the latter method is better suited to visualize a complex overall state of a system could be confirmed in informal discussions. However, a deeper verification and delineation is still pending. The potential of indexical displays and AR is great, however, we have made the experience that the effort in creating the visual metaphors, with which a real object is extended, is much greater than a symbolic representation (graphs, etc.), as they have to be tailored and can hardly be reused in other occasions. In this context, an additional potential source of error exists here, since already during the programming of the visualizations an interpretation and merging of different data points must be made in order to be able to represent an overall view. If implemented correctly, the benefit results from the reduced cognitive effort for humans observing the system.

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